Explainable Wildfire and Deforestation Detection in Satellite Imagery using CNN, Grad-CAM and SHAP

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Code and results log: github.com/arooj-zehra/xai-fire-deforestation

Abstract

We build a lightweight classifier that labels 224×224 px satellite tiles as **Normal**, **Wildfire** or **Deforestation**, and explains each decision with Grad-CAM heatmaps and SHAP attributions, served through a Streamlit dashboard. The data are 25,768 tiles drawn from two public Kaggle datasets (Canadian wildfire tiles and Planet Amazon chips). A fine-tuned MobileNetV3-Small reaches **95.0% accuracy and 0.883 macro-F1** on a held-out test set, but the rare Deforestation class is much harder (F1 0.701) than Wildfire (F1 0.986). Across a controlled grid, **data augmentation** was the only change that improved results (+0.017 macro-F1 on average over two seeds); class-weighted loss and a weighted sampler shifted the precision/recall trade-off but did not improve F1. Two baselines show the model uses real image content and also reveal that the Kaggle wildfire task is solvable largely from simple colour statistics (0.92 within-source macro-F1), so we treat the Planet task (0.817) as the honest difficulty measure. The explanations localise visible forest clearings well, but are diffuse for the Wildfire class, whose tiles show burned areas and not active fire or smoke. We report these limits explicitly.

1. Introduction and objectives

Wildfires and deforestation are increasing and are monitored largely by hand from satellite imagery. A classifier can screen tiles quickly, but authorities are unlikely to act on an alert they cannot inspect. The project therefore pairs detection with an explanation of *where* in the tile the evidence lies. The proposal set four objectives; Table 1 summarises how each was met.

Objective	Status	Evidence
1. Classify tiles as Normal / Wildfire / Deforestation	Met	Test accuracy 0.950, macro-F1 0.883 (Section 4)
2. Localise smoke, fire or cleared forest with XAI	Partly met	Cleared forest is localised; fire/smoke localisation is not demonstrated because neither dataset contains active fire or smoke (Section 5)
3. Handle class imbalance	Met (evaluated)	Weighted loss and weighted sampler compared with and without augmentation; neither improved F1 (Section 4.1)
4. Early-warning dashboard	Met	Streamlit app: upload tile, get class, confidence, heatmap (Section 6)

Table 1. Objectives and outcome.

2. Data

2.1 Sources and class definitions

Kaggle Wildfire Prediction Dataset (Satellite Images) contains 42,850 tiles of 350×350 px from Canada, labelled *wildfire* (22,710) or *nowildfire* (20,140); the wildfire tiles are locations that experienced a wildfire. **Planet “Understanding the Amazon from Space”** contains 40,479 labelled 256×256 px JPEG chips with multi-label tags (we used a public Kaggle mirror). Three classes were derived:

- **Wildfire**: Kaggle *wildfire* tiles.
- **Deforestation**: Planet tiles carrying at least one of *slash_burn* (209), *selective_logging* (340), *conventional_mine* (100), *artisinal_mine* (339) or *bare_ground* (862) tags, giving 1,768 tiles (a tile can carry several tags). The Planet tag *clear* means clear sky, not clearing, and is not used.

- **Normal**: Kaggle *nowildfire* tiles, plus Planet tiles with none of the deforestation tags. Planet tiles tagged *cloudy* (2,089) were excluded because the ground is not visible.

2.2 Why Normal comes from both sources

If Wildfire came only from Kaggle, Deforestation only from Planet and Normal from one of them, a model could score well by recognising the data source. Drawing Normal from **both** sources means that, within each source, the label is not determined by the source. Section 4.3 tests how far this protection goes.

2.3 Preparation

All images were resized to 224×224 px. To keep training light, at most 8,000 tiles were sampled per class-and-source group (random, seed 42), giving the final 25,768 tiles. A stratified 70/15/15 train/validation/test split was made on class and source (seed 42); the datasets' own splits were ignored. Truncated JPEGs in the Kaggle set are handled at load time.

Class (source)	Train	Validation	Test	Total
Normal (Kaggle)	5,600	1,200	1,200	8,000
Normal (Planet)	5,600	1,200	1,200	8,000
Wildfire (Kaggle)	5,600	1,200	1,200	8,000
Deforestation (Planet)	1,237	265	266	1,768
Total	**18,037**	**3,865**	**3,866**	**25,768**

Table 2. Dataset composition. Deforestation is 6.9% of tiles by construction, because of the per-group cap, not because it is that rare in practice.

2.4 Data-quality findings

- **Spatial shortcut in the Kaggle set.** In an earlier analysis, a k-nearest-neighbour classifier using only each tile's latitude and longitude (no image) reached 99.0-100% accuracy under several splitting schemes, because most *nowildfire* tiles were sampled near urban and farming areas and wildfire tiles in remote forest. Scores on this dataset therefore overstate real-world skill.
- **Near-duplicate check.** On 4,000 random Planet tiles, the nearest-neighbour mean absolute difference between 24×24 colour thumbnails ranged from 0.63 to a median of 3.26, with no separate cluster of true duplicates; a threshold of 3.0 flagged 21% of a 400-tile sample, which is clearly over-pruning (distinct rainforest canopy looks alike at low resolution). We used a strict threshold of 0.3; no duplicates were found among the 9,768 Planet tiles.
- **Label proxy.** Deforestation is defined through Planet's descriptive tags, which is a proxy: *bare_ground* can be natural, and farmland tiles stay Normal even though farmland is often cleared forest.

3. Method

3.1 Models and training

The main model is **MobileNetV3-Small** (about 1.5 M parameters) pre-trained on ImageNet, with its last layer replaced by a 3-way output. A **ResNet18** (about 11 M parameters) is the second architecture. Inputs are 224×224 with ImageNet normalisation. Training uses cross-entropy and Adam for 12 epochs with batch size 64: the first 2 epochs train only the classifier head (learning rate 1e-3) and the remaining 10 fine-tune all layers (1e-4). The epoch is chosen by **validation** macro-F1; the test set is scored once, with that checkpoint. Training ran in PyTorch on a Colab T4 GPU (seeds 42 and 7).

3.2 Imbalance and augmentation experiments

A 2×3 grid crossed augmentation (none vs random flips, random 90° rotations and mild colour jitter) with the imbalance strategy: plain loss, **class-weighted loss** (inverse-frequency weights: Normal 0.54, Wildfire 1.07, Deforestation 4.86) or a **weighted sampler** (classes drawn with probability inversely proportional to frequency). Metrics are accuracy, per-class precision/recall/F1, **macro-F1** (which weights the rare class equally), the confusion matrix and per-source accuracy.

3.3 Explainability

Grad-CAM is computed on the last convolutional block (a 7×7 map, upsampled to 224×224) for the predicted class. **SHAP** uses expected gradients (*GradientExplainer*) with 50 background training tiles and 50 samples per tile, giving a signed attribution per pixel and colour channel. As a **faithfulness check**, the 10 most important of 49 Grad-CAM cells are replaced by the mean colour and the drop in the predicted-class probability is compared with masking 10 random cells, on 60 correctly classified test tiles.

3.4 Baselines for the shortcut analysis

Two trivial predictors are scored on the same test tiles: *source-only* (predict the commonest class of the tile's data source) and *colour-statistics* (gradient boosting on 8 numbers per tile: mean and standard deviation of each RGB channel, and the 10th and 90th brightness percentiles).

4. Results

4.1 Imbalance and augmentation grid

Table 3 ranks the runs by validation macro-F1, the pre-agreed selection rule. **Augmentation alone (run B)** was best on validation and on test.

Run	Val macro-F1	Test acc.	Test macro-F1	Defo. precision	Defo. recall	Defo. F1	Planet acc.
B: augmentation	**0.891**	**0.950**	**0.883**	**0.696**	**0.707**	**0.701**	**0.891**
G: ResNet18 + aug.	0.880	0.943	0.874	0.621	0.763	0.685	0.872
A: baseline (no aug.)	0.880	0.943	0.868	0.640	0.688	0.663	0.872
F: sampler + aug.	0.878	0.936	0.864	0.570	0.778	0.658	0.853
E: sampler	0.876	0.946	0.865	0.700	0.613	0.653	0.882
C: weighted loss	0.872	0.937	0.859	0.612	0.688	0.648	0.864
D: weighted loss + aug.	0.862	0.928	0.860	0.529	0.861	0.655	0.836

Table 3. Seed-42 runs, MobileNetV3-Small except G. Defo. = Deforestation class. Test set: 3,866 tiles (266 Deforestation).

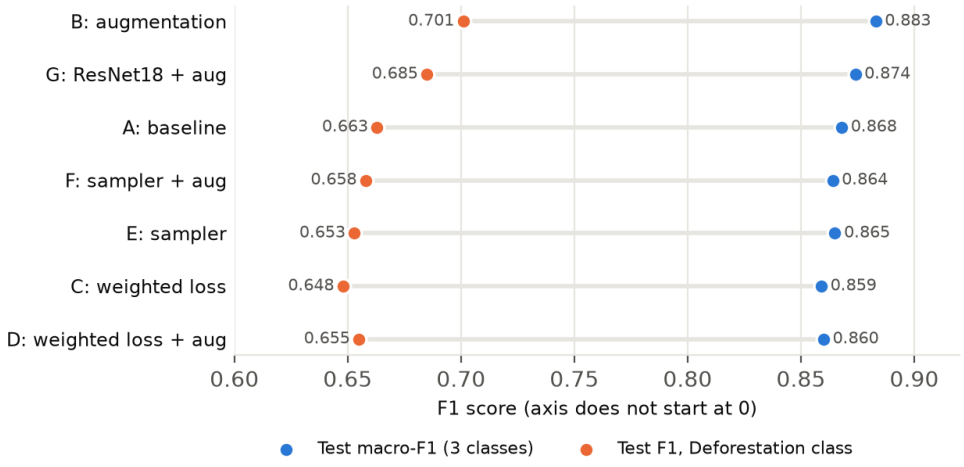


Figure 1. Test macro-F1 (blue) and Deforestation F1 (orange) per run, ordered by validation macro-F1.

Is the augmentation gain real? Re-running the baseline and run B with a second seed (7) gave the same direction:

	Seed 42	Seed 7	Mean
Baseline - test macro-F1	0.868	0.859	0.864
Augmentation - test macro-F1	0.883	0.879	0.881
Baseline - Deforestation F1	0.663	0.646	0.655
Augmentation - Deforestation F1	0.701	0.688	0.695
Baseline - Planet accuracy	0.872	0.876	0.874
Augmentation - Planet accuracy	0.891	0.892	0.892

Table 4. Two-seed check. The seed ranges for the two settings do not overlap, but two seeds is a small sample.

- Augmentation also reduced overfitting: final training loss 0.078 against 0.012 for the baseline, and validation loss stayed near 0.12 instead of rising to about 0.21.
- Weighted loss and the sampler changed the operating point without improving F1: run D catches 86% of deforestation tiles but with 53% precision, while run E has 70% precision and 61% recall. For an early-warning use, where a missed event costs more than a false alarm, the high-recall setting is a defensible choice.
- ResNet18 with augmentation (0.874) did not beat MobileNetV3-Small (0.881 on average) and is about 7 times larger and slower per epoch (about 85 s against 70 s with augmentation on a T4), so MobileNetV3-Small with augmentation is the final model.

4.2 Final model

Class	Precision	Recall	F1	Test tiles
Normal	0.959	0.960	0.960	2,400
Wildfire	0.989	0.983	0.986	1,200
Deforestation	0.696	0.707	0.701	266
Overall	**accuracy 0.950**		**macro-F1 0.883**	**3,866**

Table 5. Final model (run B) on the test set. Accuracy by source: Kaggle 0.986, Planet 0.891.

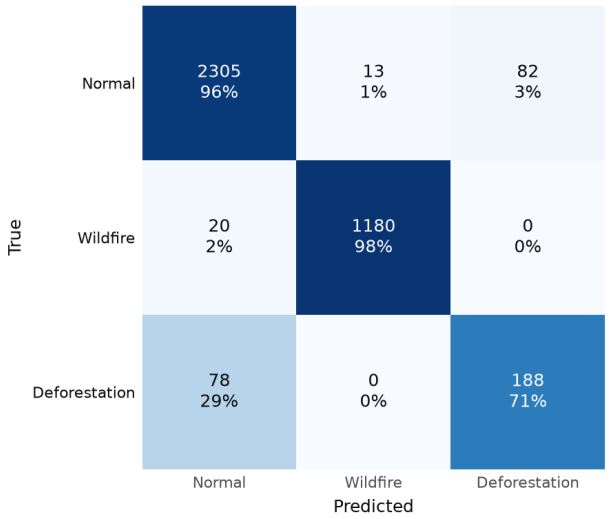


Figure 2. Test confusion matrix of the final model (counts and row percentages). Every Deforestation error is a confusion with Normal.

4.3 Does the model rely on shortcuts?

Predictor	Accuracy	Macro-F1	Defo. F1	Kaggle within-source macro-F1	Planet within-source macro-F1
CNN (final model)	**0.950**	**0.883**	**0.701**	**0.986**	**0.817**
Colour-statistics boosting	0.890	0.749	0.413	0.923	0.659
Source-only	0.621	0.429	0.000	0.333	0.450

Table 6. Final model against trivial baselines on the same 3,866 test tiles. Within-source macro-F1 scores each dataset's own task (Kaggle: Normal vs Wildfire; Planet: Normal vs Deforestation).

- The model uses image content: it beats the colour-statistics baseline by 0.13 macro-F1 overall and by 0.16 within Planet, and the source-only baseline never finds Deforestation.
- The Kaggle task is easy: eight colour numbers already give 0.92, consistent with the spatial shortcut in Section 2.4. The Planet task (0.817) is the honest difficulty measure.
- The model never confuses classes across sources (no Wildfire predictions on Planet tiles, no Deforestation predictions on Kaggle tiles), so it effectively solves two binary problems and can recognise the source. Drawing Normal from both sources ensures the label is not determined by the source *within* each dataset, but does not remove source information.

5. Explainability results

Figure 3 (below) shows tiles from each class with Grad-CAM (red = most influential region) and SHAP (red supports, blue opposes the predicted class). The tiles were chosen by seeded random sampling (two correct tiles and one error per class), not hand-picked, so they are a fair sample of behaviour.

5.1 What the explanations show

- **Deforestation: localised and plausible.** A small pale clearing inside dark forest (Figure 3, tile 7) is picked out by both Grad-CAM and SHAP, and the cleared brown corner in tile 8 is highlighted.
- **Wildfire: diffuse.** Heatmaps spread across large areas (roads, field strips, forest texture) with no identifiable fire, smoke or burn scar. The Kaggle wildfire tiles show areas that burned at some point, not active fire, so there is little localised evidence to highlight; the model appears to rely on broad context (dense remote forest versus farmland or suburbs).
- **Errors often look like label noise.** A Planet “Deforestation” tile predicted Normal at 0.98 (tile 9) shows plain dark forest with no visible clearing; a Normal tile predicted Deforestation (tile 3) shows pale clearings. A Wildfire tile of fields and a road (tile 6) is read as Normal.
- **No colour-channel story.** The share of SHAP importance per channel is nearly identical across classes (Figure 4), so channel importance reflects general sensitivity, not class-specific colour use.

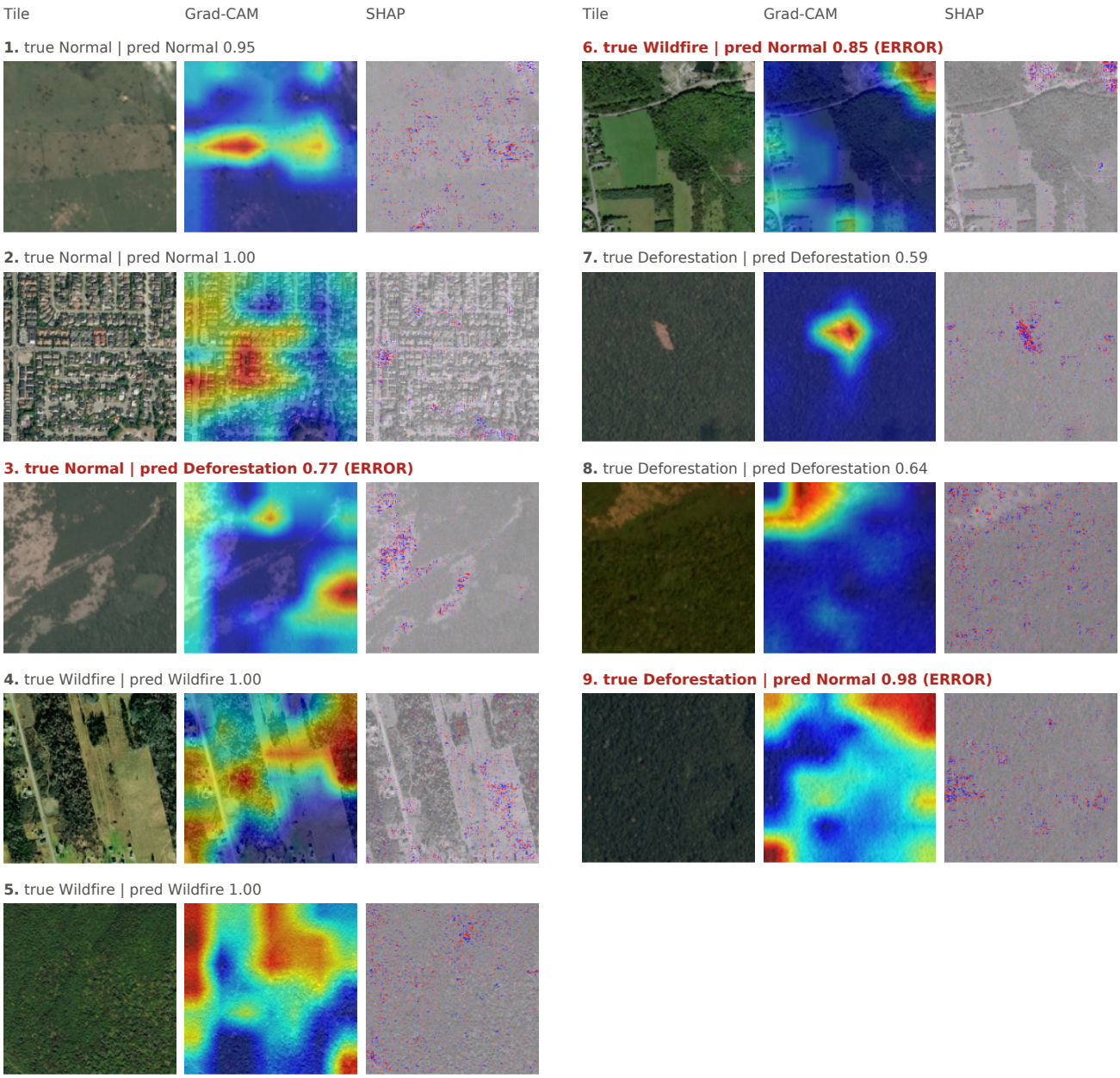


Figure 3. Tile, Grad-CAM and SHAP for correct predictions and errors (red titles). Tiles 1-3 are Normal, 4-6 Wildfire and 7-9 Deforestation.

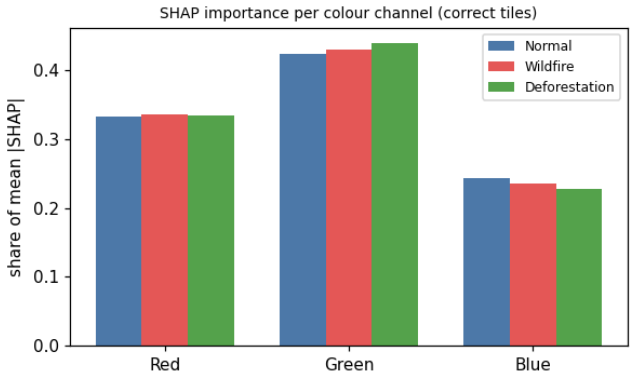


Figure 4. Share of mean absolute SHAP per colour channel, by predicted class (correct tiles).

5.2 Faithfulness

Class	Confidence drop: top Grad-CAM cells	Confidence drop: random cells
All (60 tiles)	0.204	0.131
Normal	0.055	0.022
Wildfire	0.055	0.012
Deforestation	0.501	0.357

Table 7. Drop in predicted-class probability when 10 of 49 cells are masked. Grad-CAM cells beat random cells on 62% of tiles.

Grad-CAM is meaningfully faithful for Deforestation, where hiding its cells costs far more confidence than hiding random ones. For Normal and Wildfire the model barely reacts to losing 20% of the tile, which agrees with the picture of decisions driven by global context. Overall the evidence is moderate: the 62% win rate on 60 tiles is only borderline significant, and the explanations were not validated by remote-sensing experts. A synthetic sanity test (tiles with a planted colour patch) confirmed that the implementation highlights a known region, so the diffuse Wildfire heatmaps reflect the model and data, not a bug.

6. Dashboard

The Streamlit app (*app/app.py*) loads the trained checkpoint on CPU. The user uploads a tile, or loads a random test tile, and receives: an **alert banner** (alert when a non-Normal class is predicted with at least an adjustable confidence threshold, a softer notice below it, or no alert); the class probabilities; the **Grad-CAM** overlay for any chosen class; and an optional **SHAP** overlay, which is off by default because it is much slower. A sidebar note states the model's limits. A map view (Folium) was left out to keep the scope fixed.

7. Limitations

- **Per-tile classification, not change detection.** The tool scores single images; it does not use time series, so “early warning” means screening tiles, not forecasting.
- **The Wildfire class is not active fire.** The Kaggle tiles are historical burn locations; results say nothing about detecting smoke or flames, and the dataset's spatial shortcut inflates scores (Section 4.3).
- **Deforestation labels are a tag-based proxy** with visible noise (Section 5.1), which caps achievable F1 on that class.
- **Class balance is artificial.** The per-group cap makes Deforestation 6.9% of the data. At real-world prevalence, precision would be lower and false alarms would dominate; thresholds would need recalibration.
- **Evaluation is a single random split.** One test set was shared by all experiments, only two seeds were run, and tiles cut from the same scene may fall in different splits, which can inflate scores. No confidence intervals were computed; differences below about 0.02 macro-F1 should not be over-interpreted.
- **Explanation quality is bounded:** Grad-CAM has a coarse 7×7 grid, SHAP used limited samples, the faithfulness test covers 60 tiles, and no expert validation was done.
- **Domain:** trained at 224×224 px on two sensors and regions; it is untested elsewhere.

8. Future work

- Add expert-annotated burn masks (for example Copernicus EMS Sentinel-2 data) for spatially clean labels and true localisation targets.
- Evaluate with spatially blocked splits to remove the scene and location leakage, and add more seeds with confidence intervals.
- Use data with active fire or smoke (for example thermal or shortwave-infrared bands) so that fire and smoke localisation can be tested.

- Select the alert threshold from the cost of missed events versus false alarms and recalibrate for real prevalence; consider the high-recall variant (run D).
- Add temporal change detection and a map view.

9. Reproducibility

Everything is in the repository. Environment: Python 3.12, PyTorch 2.14.1, torchvision 0.29.1, SHAP 0.54.1, Streamlit 1.66.1 (pinned in *requirements.lock.txt*). Pipeline: *scripts/build_dataset.py* (data), *python -m src.train* (training; *--aug*, *--imbalance*, *--model*, *--seed*), *scripts/summarize_runs.py* (results table), *scripts/source_report.py* (shortcut analysis), *python -m src.run_explain* (explanations), *streamlit run app/app.py* (dashboard). All numbers in this report come from the logs in *docs/RESULTS.md*; the data and checkpoints are not stored in git.

References

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